

Ryan Walker

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Education

Georgia Institute of Technology

Bachelor of Science in Computer Science

Atlanta, GA

Expected May 2028

GPA: 4.00 / 4.00

Study Abroad: Georgia Tech Europe, Metz, France — May–Aug 2025.

Relevant Coursework: Data Structures & Algorithms, Advanced Algorithms & Complexity, Object-Oriented Programming, Objects & Design, Computer Organization & Programming, Systems & Networks, Networking I, Intro to AI, Perception & Robotics, Linear Algebra, Probability & Statistics, Multivariable Calculus, Differential Equations, Discrete Math, Applied Combinatorics.

Technical Skills

Languages: Java, Python, C, Assembly, JavaScript, SQL, MATLAB, HTML/CSS, Pine Script

Frameworks & Libraries: PyTorch, NumPy, SciPy, Matplotlib, FAISS, Django, Flask, React, Streamlit, JUnit, Chart.js, YFinance

Tools & Platforms: AWS, Docker, Terraform, CI/CD, Datadog, Observe, Git/GitHub, Claude Code, Gemini CLI, Supabase, Vercel, PythonAnywhere, Google OR-Tools, Google Colab, L^AT_EX

Experience

Sonatype — Associate Software Engineer (Part-Time)

Remote · Aug 2026 – Present

- Retained part-time through the academic year following the summer internship, continuing to deliver Lifecycle product features, AWS/Terraform infrastructure work, and unit-tested code alongside the SCA team.

Sonatype — Associate Software Engineering Intern

Remote · May 2026 – Aug 2026

- Contributed to the Software Composition Analysis (SCA) team, delivering improvements to the Lifecycle product and AWS infrastructure within a sprint-based Agile environment, writing unit tests to validate each change.
- Played a key role in migrating cloud infrastructure from EKS to ECS Fargate, enhancing the scalability and efficiency of the Lifecycle product.
- Managed AWS resources with Terraform and Docker across CI/CD deployment pipelines — deploying, managing, and destroying containerized product instances on ECS to enable robust cloud testing across the SDLC.
- Assisted the transition from Datadog to the Observe platform, migrating and configuring essential monitors and charts for analytics and monitoring.

Personal Portfolio Management — Quantitative Analyst / Developer

Remote · May 2020 – Present

- Developed and hosted a Streamlit risk dashboard — later migrated to a React app on Vercel — using a Supabase Postgres database and FintHub/YFinance APIs for real-time portfolio concentration (HHI), risk-by-expiration, projections, and Monte Carlo simulations.
- Conducted research and backtesting of options strategies, including call credit spreads, put debit spreads, iron condors, and the wheel.
- Ran live paper tests of the wheel over a small-cap universe, with entry, exit, and screening rules from a 104-name, 6-year backtest.
- Used dashboard analytics to optimize position sizing and flag over-leveraged positions, driving risk mitigation and allocation decisions.

Georgia Tech Investments Committee — Junior Analyst

Atlanta, GA · May 2025 – Present

- Completed a rigorous analyst mentorship program for the Georgia Tech Investments Committee, which manages a \$2.7M endowment.
- Served two semesters as a Junior Analyst in the Media & Telecom sector, conducting in-depth company and industry research.
- Developed comprehensive primers and pitches using DCF and relative-valuation models, directly leading to trimming one existing holding and acquiring a new one.
- Re-evaluated investment theses for current holdings, applying business valuation and macro analysis to inform strategic decisions.

Projects

Image Classification CNN — PyTorch, VGG16, Matplotlib

Independent Course Project

- Developed a high-performance CNN in PyTorch classifying 75 distinct butterfly species with 89–90% accuracy on a test dataset.
- Applied transfer-learning techniques by leveraging the VGG16 model and retraining two fully connected layers on a 3,000-image dataset.
- Generated training-performance graphs with Matplotlib to monitor model behavior and prevent overfitting throughout the training process.

Fitness Tracking Web Application — Django, JavaScript, Chart.js

5-Person Team Project

- Developed a full-stack fitness tracking web application as part of a 5-person team using Django, JavaScript, CSS, and HTML.
- Engineered and integrated the Gemini API to generate user-specific workout plans based on individual goals and biometric inputs.
- Designed and implemented dynamic dashboard data visualizations with Chart.js, enabling users to track fitness metrics, goals, and progress.
- Contributed core features including user authentication, workout logging, goal management, and a social networking component; deployed to production on PythonAnywhere.

Leadership

Theta Xi Fraternity — Social Chair

Atlanta, GA · Oct 2024 – Present

- Managed an ~\$18K per-semester budget and led a team of assistants to plan and execute large-scale social events with attendance ranging from 80 to 800 people.
- Orchestrated 5–6 date nights, 2–3 parties, and up to 8 mixers each semester, consistently achieving higher engagement than previous years, and fostered inter-fraternity relations through joint events.